

SPECIFICATION

PROFENOFOS TECHNICAL

SECTION 1: CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

PRODUCT NAME	:	Profenofos Technical
CHEMICAL NAME	:	O – (4 – bromo – 2 – chlorophenyl) – O – ethyl – S – n –propyl Phosphorothioate
CHEMICAL FAMILY	:	Insecticide.
SYNONYMS	:	-----
MANUFACTURER	:	Meghmani Organics Limited (Corporate Office) "Meghmani House" Behind Safal Profitaire, Corporate Rd., Prahladnagar Ahmedabad – 380 015. Gujarat, India.

FOR MORE INFORMATION CALL AT FACTORY:

Tel: 91-02641-291017, 91-99099 95945.

Fax: -----

AT CORPORATE OFFICE:

Tel: 91 79-7176 1000, 2970 9600

Fax: 91-79-2970 9605

SECTION 2: COMPOSITION, INFORMATION OF INGREDIENTS

COMPONENT	PERCENTAGE	CAS NUMBER	HAZARD DATA
Profenofos Technical	95% min.	41198 – 08 – 7	Presence of no Specific

SECTION 3: HAZARDS IDENTIFICATION SUMMARY

Symptoms of Acute Exposure

This product is a cholinesterase inhibitor. Principal routes of exposure are skin contact and inhalation. Symptoms of cholinesterase inhibition may include salivation, sweating, headache, nausea, muscle twitching, tremors, poor coordination, blurred vision, tears, abdominal cramps, and diarrhea and chest discomfort.

Irritating to eyes, respiratory system and skin. May cause sensitization by skin contact. Vapors may cause drowsiness and dizziness.

Hazardous Decomposition Products : Can decompose at high temperatures forming toxic gases.

Physical Properties : Appearance : Amber liquid
Odor : Pungent, garlic-like odor

Unusual Fire, Explosion and Reactivity Hazards

Combustible liquid. Can release vapors that form explosive mixtures at temperatures at or above the flash point. Heavy vapors can flow along surfaces to distant ignition sources and flash back. During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

SECTION 4: FIRST AID MEASURES

Have the product container, label or Material Safety Data Sheet with you when calling doctor, or going for treatment.

Ingestion : If swallowed: doctor immediately for treatment advice. Do not give any liquid to the person. Do not induce vomiting unless told to do so after doctor. Do not give anything by mouth to an unconscious person.

Eye Contact : If in eyes: Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after 5 minutes, then continue rinsing eye. Call doctor for treatment advice.

Skin Contact : If on skin or clothing: Take off contaminated clothing, Rinse skin immediately with plenty of water for 15-20 minutes. Call doctor for treatment advice.

Inhalation : If inhaled: Move person to fresh air. If person is not breathing, call an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call doctor for further treatment advice.

Notes to Physician

Persons suffering a temporary allergic reaction may respond to treatment with antihistamines or steroid creams and/or systemic steroids.

Contains petroleum distillate – vomiting may cause aspiration pneumonia.

Contains an organophosphate that inhibits cholinesterase. Consider taking venous blood for determination of blood cholinesterase activity (use heparin tube). Administer atropine sulfate, either by intramuscular or intravenous injection, dependent on severity of poisoning. Specific antidotes are oximes such as Pralidoxime (2-PAM). Avoid the use of opiates and phenothiazine. Carefully monitor patient because continued absorption of the poison may occur resulting in fatal relapse after initial improvement.

Medical Condition Likely to be Aggravated by Exposure : None known.

SECTION 5: FIRE FIGHTING MEASURES

Fire and Explosion

Flash Point (Test Method) : 133°F (Setaflash CC)
Flammable Limits (% in Air) : Lower : 0.9%, Upper : 5.2%
Autoignition Temperature : 670°F
Flammability : Combustible liquid

Unusual Fire, Explosion and Reactivity Hazards

Combustible liquid. Can release vapors that form explosive mixtures at temperatures at or above the flash point. Heavy vapors can flow along surfaces to distant ignition sources and flash back. During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

In Case of Fire

Use appropriate extinguishing media for combustibles in the area. Wear full protective clothing and self-contained breathing apparatus. Evacuate nonessential personnel from the area to prevent human exposure to fire, smoke, fumes or products of combustion. Prevent use of contaminated buildings, area, and equipment until decontaminated. Water runoff can cause environmental damage. If water is used to fight fire, dike and collect runoff.

SECTION 6: ACCIDENTAL RELEASE MEASURES

In Case of Spill or Leak

Control the spill at its source. Contain the spill to prevent from spreading or contaminating soil or from entering sewage and drainage systems or any body of water. Clean up spills immediately, observing precautions outlined in Section 8. Cover entire spill with absorbing material and place into compatible disposal container. Scrub area with hard water detergent (e.g. commercial products such as Tide, Joy, Spic and Span). Pick up wash liquid with additional absorbent and place into compatible disposal container. Once all material is cleaned up and placed in a disposal container, seal container and arrange for disposition.

SECTION 7: HANDLING AND STORAGE

Store the material in a well-ventilated, secure area out of reach of children and domestic animals. Do not store food, beverages or tobacco products in the storage area. Prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling.

SECTION 8: EXPOSURE CONTROLS, PERSONAL PROTECTION

THE FOLLOWING RECOMMENDATIONS FOR EXPOSURE CONTROLS/PERSONAL PROTECTION ARE INTENDED FOR THE MANUFACTURE, FORMULATION, PACKAGING AND USE OF THIS PRODUCT. FOR COMMERCIAL APPLICATIONS AND/OR ON-FARM APPLICATIONS CONSULT THE PRODUCT LABEL.

- Ingestion** : Prevent eating, drinking, tobacco usage and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling.
- Eye Contact** : Where eye contact is likely, use chemical splash goggles. Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.
- Skin Contact** : Where contact is likely, wear chemical-resistant (such as nitrile or butyl) gloves, coveralls, socks and chemical-resistant footwear. For overhead exposure, wear chemical-resistant headgear.
- Inhalation** : A respirator is not normally required when handling this substance. Use effective engineering controls to comply with occupational exposure limits.

In case of emergency spills : Use a NIOSH approved respirator with any R, P or HE filter.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Appearance	:	Amber liquid with Pungent, garlic-like odor
Physical state	:	Liquid
Colour	:	Amber
Odor	:	Pungent, garlic-like odor
Melting Point	:	Not Applicable
Boiling Point	:	500.7 ± 0.3 K at 760 mm of Hg
Specific Gravity/Density	:	1.18g/cm ³ @ 68°F (20°C)
pH	:	3 – 5 (1% solution in water)
Solubility in water	:	In buffer water (pH 5.01) was 0.0218 ± 0.0002 g/L at 20 ± 1 °C In buffer water (pH 7.06) was 0.0226 ± 0.0004 g/L at 20 ± 1 °C.
Solubility in Organic Solvents	:	Profenofos technical is highly miscible in below tested organic solvents by CIPAC MT 181 Method Acetone, Methanol, Ethyl Acetate, 1,2-Dichloroethane n-Heptane : > 250 g/L p-Xylene : >200 g/L
Vapor Pressure	:	3.89944 x 10 ⁻⁴ Pa @ 25 °C.
Partition Coefficient	:	K _{OW} logP = 4.39 ± 0.02 at 20 ± 1 °C.

SECTION 10: STABILITY AND REACTIVITY

Stability	:	Stable under normal use and storage conditions.
Hazardous Polymerization	:	Will not occur,
Conditions to Avoid	:	Flame, heat, ignition sources. Can decompose at high temperatures forming toxic gases.
Hazardous Decomposition Products:		– NA –
Materials to Avoid	:	Strong oxidizers.

SECTION 11: TOXICOLOGICAL INFORMATION

ACUTE TOXICITY	:	
Acute Oral LD ₅₀ Rat	–	>500 mg/kg.
Acute Dermal LD ₅₀ Rat	–	>2000 mg/kg
Acute Inhalation LC ₅₀ Rat	–	> 5.00 mg/l air
Primary Skin irritation Study to Rabbit	–	Non Irritant
Acute Eye Irritation to Rabbit	–	Non Irritant
Skin Sensitization to Guinea Pigs	–	Non Sensitizer

<u>Reproductive/Developmental Effects</u>	:	Profenofos	:	None observed
<u>Chronic/Subchronic Toxicity Studies</u>	:	Profenofos	:	Cholinesterase inhibition.
<u>Carcinogenicity</u>	:	Profenofos	:	None observed.

SECTION 12: ECOLOGICAL INFORMATION

Summary of Effects

Profenofos Very toxic to aquatic organisms; may cause long-term adverse effects in the aquatic environment. Toxic to birds. Very toxic to honey bees.

Eco-Acute Toxicity

Profenofos	Bees LC ₅₀ /EC ₅₀ 0.1 ug/bee
	Invertebrates (Water Flea) LC ₅₀ /EC ₅₀ 0.0014 ppm
	Fish (Trout) LC ₅₀ /EC ₅₀ 0.025 ppm
	Fish (Bluegill) LC ₅₀ /EC ₅₀ 0.3 ppm
	Birds (8 - day dietary - Bobwhite Quail) LC ₅₀ /EC ₅₀ 57 ppm
	Birds (8 - day dietary - Mallard Duck) LC ₅₀ /EC ₅₀ 1,647 ppm

Eco-Chronic Toxicity

Profenofos	:	Mallard Duck 8-days LC ₅₀ 150 ppm
		Bobwhite Quail 8-days LC ₅₀ 70 ppm
		Japanese Quail 8-days LC ₅₀ > 1,000 ppm

Environmental Fate

Profenofos	:	The information presented here is for the active ingredient,
Profenofos	:	Low bioaccumulation potential. Not persistent in soil or water. Low mobility in soil. Sinks in water (after 24 hr.)

SECTION 13: DISPOSAL CONSIDERATIONS

Disposal	:	Do not reuse product containers. Dispose of product containers, waste containers, and residues according to local, state, and federal health and environmental regulations.
Characteristic Waste:		Ignitable D001

SECTION 14: TRANSPORT INFORMATION

Proper Shipping Name : Organophosphorus Pesticide, Liquid Toxic, N.O.S.
(Profenofos Technical)
Class : 6.1
UN Number : 3018
Packing Group : III

SECTION 15: REGULATORY INFORMATION

CAS No. : [41198 – 08 – 7]
Component name : Profenophos

PRECAUTIONS

Wash hands and exposed skin after use.
Wash splashes from skin or eyes immediately with plenty of water.

SECTION 16: OTHER INFORMATION

The above information is believed to be correct but does not purport to be all inclusive and shall be used only as a guide. The information in this document is based on the present state of our knowledge and is applicable to the product with regard to appropriate safety precautions. It does not represent any guarantee of the properties of the product. Meghmani shall not be held liable for any damage resulting from handling or from contact with the above product.